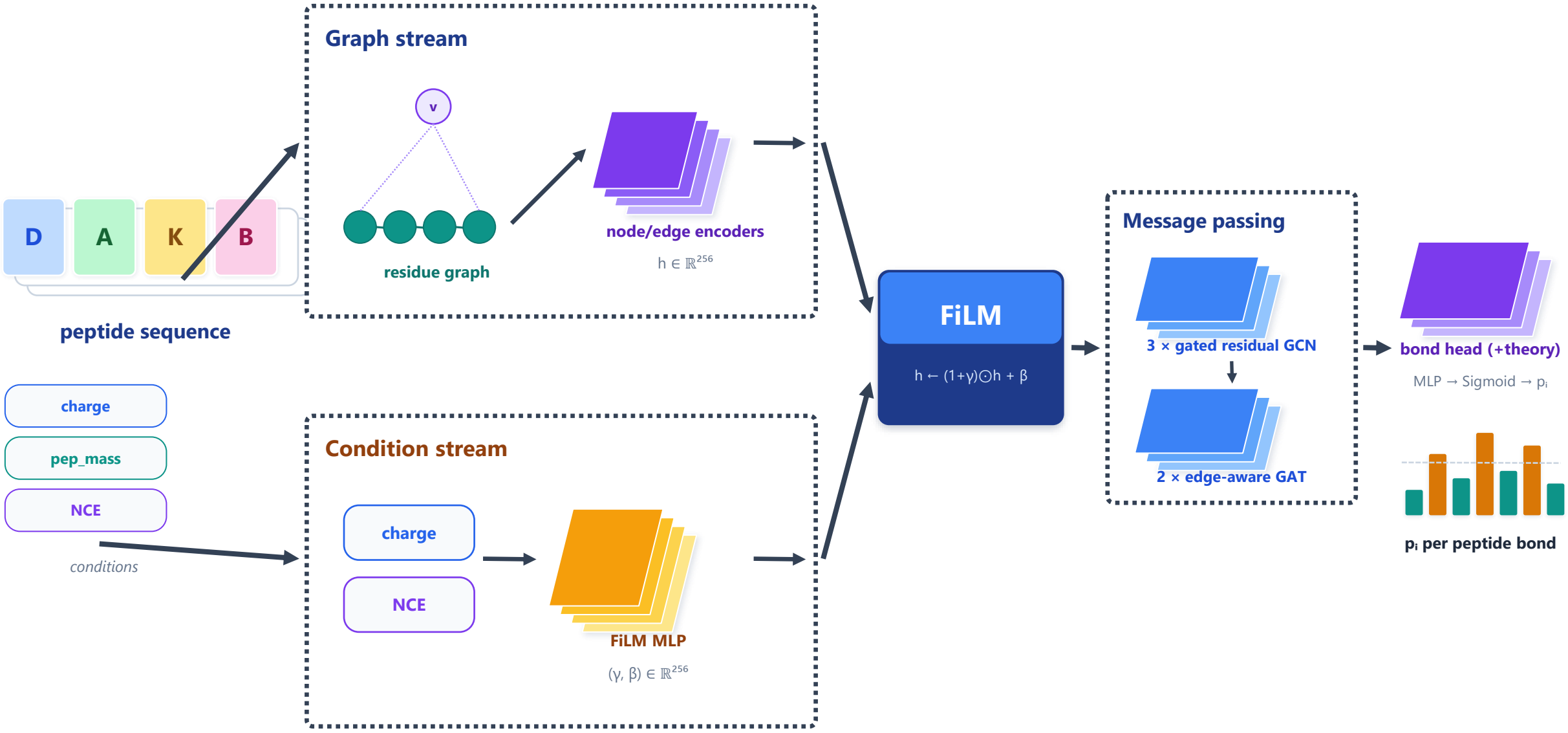

Graph stream
v
residue graph
node/edge encoders
h ∈ ℝ²⁵⁶
D
A
K
B
peptide sequence
charge
pep_mass
NCE
conditions
Condition stream
charge
NCE
FiLM MLP
(γ, β) ∈ ℝ²⁵⁶
FiLM
h ← (1+γ)⊙h + β
Message passing
3 × gated residual GCN
2 × edge-aware GAT
bond head (+theory)
MLP → Sigmoid → pᵢ
pᵢ per peptide bond